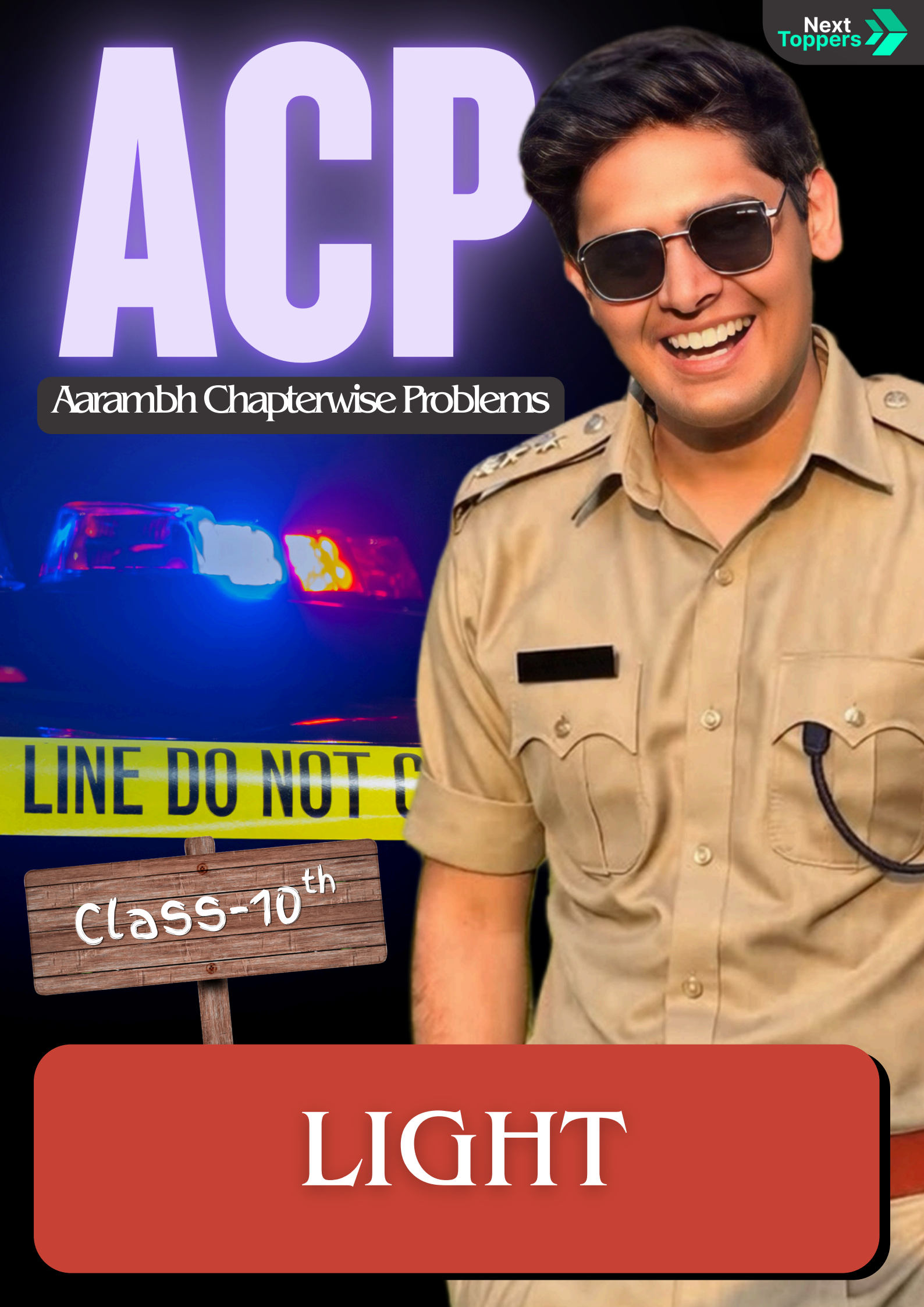


ACP

Aarambh Chapterwise Problems



LINE DO NOT CROSS

Class-10th

LIGHT

LIGHT

Read the question properly!

PART 1 (Basic MCQs)

1. When a plane mirror is rotated through a certain angle, the reflected ray turns through twice as much and the size of the image:

- (a) is doubled
- (b) is halved
- (c) becomes infinite
- (d) remains the same

2. If an object is placed symmetrically between two plane mirrors, inclined at an angle of 72 degrees, then the total no. of images formed is:

- (a) 5
- (b) 4
- (c) 2
- (d) infinite

3. Which statement is true for the reflection of light?

- (a) The angle of incidence and reflection are equal.
- (b) The reflected light is less bright than the incident light.
- (c) The sum of the angle of incidence and reflection is always greater than 90° .
- (d) The beams of the incident light, after reflection, diverge at unequal angles.

**JOSH
METER**



4. The focal length of a plane mirror is:

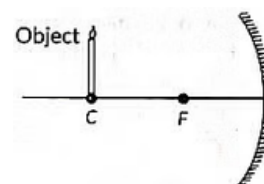
- (a) 0
- (b) infinite
- (c) 25 cm
- (d) -25 cm

5. A ray of light passes through the optical centre of a convex lens. How does the ray emerge after refraction?

- (a) It bends towards the principal axis
- (b) It bends away from the principal axis
- (c) It passes undeviated
- (d) It gets reflected back

6. Which of the following statements is not true in reference to the diagram shown above?

- (a) Image formed is real.
- (b) Image formed is enlarged.
- (c) Image is formed at a distance equal to double the focal length.
- (d) Image formed is inverted.



7. An object is placed at a distance of 40 cm in front of a concave mirror of a focal length of 20 cm. The image produced is:

- (a) virtual and inverted

- (b) real and erect
- (c) real, inverted and of the opposite size as that of the object
- (d) real, inverted and of the same size as that of the object

8. A student conducts an experiment using a convex lens. He places the object at a distance of 60 cm in front of the lens and observes that the image is formed at a distance of 30 cm behind the lens. What is the power of the lens?

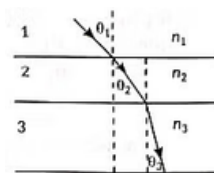
- (a) 0.005 dioptre
- (b) 0.05 dioptre
- (c) 5 dioptre
- (d) 50 dioptre

9. An image of an object produced on a screen is about 36 cm using a convex lens. The image produced is about 3 times the size of the object. What is the size of the object?

- (a) 12 cm
- (b) 33 cm
- (c) 39 cm
- (d) 108 cm

10. In the diagram shown above n_1 , n_2 and n_3 are refractive indices of the media 1, 2 and 3 respectively. Which one of the following is true in this case.

- (a) $n_1 = n_2$
- (b) $n_1 > n_2$
- (c) $n_2 > n_3$
- (d) $n_3 > n_1$



11. A student studies that a convex mirror always forms a virtual image irrespective of its position. What causes the convex mirror to always form a virtual image?

- (a) Because the reflected ray never intersects
- (b) Because the reflected ray converges at a single point
- (c) Because the incident ray traces its path back along the principal axis
- (d) Because the incident ray of a convex mirror gets absorbed in the mirror

12. A 10 mm long awl pin is placed vertically in front of a concave mirror at a distance of 60 cm. Its image is formed at 30 cm in front of the mirror. The focal length of this mirror is:

- (a) -30 cm
- (b) -20 cm
- (c) -40 cm
- (d) -60 cm

13. Rahul conducts an experiment using an object of height 10 cm and a concave lens with a focal length of 20 cm. The object is placed at a distance of 25 cm from the lens. Can the image be formed on a screen?

- (a) Yes, as the image formed will be real
- (b) Yes, as the image formed will be erect
- (c) No, as the image formed will be virtual
- (d) No, as the image formed will be inverted

14. Magnification produced by a rearview mirror fitted in vehicles:

- (a) is less than one
- (b) is more than one

- (c) is equal to one
- (d) can be more than or less than one, depending upon the position of the object in front of it

15. A student conducts an activity using a concave mirror with a focal length of 10 cm. He placed the object 15 cm from the mirror. Where is the image likely to form?

- (a) At 6 cm behind the mirror
- (b) At 30 cm behind the mirror
- (c) At 6 cm in front of the mirror
- (d) At 30 cm in front of the mirror

16. The image of an object placed in front of a convex mirror is formed at:

- (a) the object itself
- (b) twice the distance of the object in front of the mirror
- (c) half the distance of the object in front of the mirror
- (d) behind the mirror

17. A full-length image of a distant tall building can definitely be seen using:

- (a) a concave mirror
- (b) a convex mirror
- (c) a plane mirror
- (d) both concave as well as plane mirrors

18. A student conducts an activity using a flask of height 15 cm and a concave mirror. He finds that the image formed is 45 cm in height. What is the magnification of the image?

- (a) -3 times
- (b) $-1/3$ times
- (c) $1/3$ times
- (d) 3 times

19. Which of the following can make a parallel beam of light from a point source incident on it?

- (a) Concave mirror as well as convex lens
- (b) Convex mirror as well as concave lens
- (c) Two plane mirrors placed at 90° to each other
- (d) Concave mirror as well as concave lens

20. A student studies that the speed of light in air is 300000 km/sec, whereas the speed in a glass slab is about 197000 km/sec. What causes the difference in the speed of light in these two media?

- (a) Difference in density
- (b) Difference in temperature
- (c) Difference in the amount of light
- (d) Difference in the direction of wind flow

ASSERTION AND REASON TYPE QUESTION (1 Marks)



iyaha Marks katate hi

In the following questions a statement of Assertion is followed by a statement of Reason.

Mark the correct choice as two statements are given one labeled Assertion (A) and the other labeled Reason (R).

Select the correct answer to these questions from the codes (a), (b), (c) and (d) as given below:

- (a) Both A and R are true, and R is correct explanation of the assertion.
- (b) Both A and R are true, but R is not the correct explanation of the assertion.

32. "The refractive index of carbon disulphide is 1.63." What is the meaning of this statement in relation to speed of light?
33. The outer surface of a hollow sphere of aluminium of radius 50 cm is to be used as a mirror. What will be the focal length of this mirror? Which type of spherical mirror will it provide?
34. A girl was playing with a thin beam of light from her laser torch by directing it from different directions on a convex lens held vertically. She was surprised to see that in a particular direction the beam of light continues to move along the same direction after passing through the lens. State the reason for this observation.
35. A ray of light enters a rectangular glass slab of refractive index 1.5. It is found that the ray emerges from the opposite face of the slab without being displaced. If its speed in air is 3×10^8 m/s then what is its speed in glass?
36. A student places an object between the pole and focus of a concave mirror and observes that the image formed is erect and enlarged. Another student claims that all images formed by concave mirrors are inverted. Who is correct? Justify your answer with reason.
37. A driver prefers using a convex mirror as a rear-view mirror instead of a plane mirror. Explain the scientific reason behind this choice.
38. A concave mirror forms an erect and magnified image of an object. Where is the object placed with respect to the mirror? Explain.
39. A student says that a real image can never be formed by a convex lens. Do you agree? Justify your answer with reasoning.
- Q40. Why does a ray of light bend towards the normal when it enters glass from air, but bend away from the normal when it comes out of glass into air?
41. An object is placed very close to a convex lens and the image formed is virtual. Explain why the image cannot be obtained on a screen.

LONG ANSWER TYPE QUESTIONS (5 Marks)

(pahle points socho firr likho)

42. A concave mirror forms a real image three times larger than the object. If the image is formed 36 cm in front of the mirror, calculate the position of the object and the focal length of the mirror.
43. An object is placed between the focus and centre of curvature of a concave mirror. State the nature, size, and position of the image formed. How will the image change if the object is moved towards the focus?
44. A convex lens forms a real image of an object on a screen placed 80 cm away from the lens. If the image is twice the size of the object, calculate the focal length of the lens.
45. A student uses a concave mirror of focal length 20 cm to obtain an image of an object placed at different positions. At what position should the object be placed so that the image formed is highly enlarged and virtual? Give reason.
46. Why does a ray of light bend away from the normal when it travels from glass to air? State the reason.

47. A convex lens and a concave lens have the same focal length. Compare their behaviour when parallel rays of light fall on them. Also state the type of image formed by each lens.

48. An object placed 24 cm from a concave lens produces a virtual image 8 cm from the lens on the same side as the object. Calculate the focal length and power of the lens.

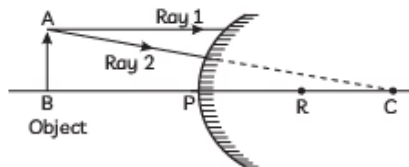
49. A rear-view mirror in a vehicle forms an image of magnification $+0.25$. Identify the type of mirror used and explain why this mirror is preferred over a plane mirror.

50. A student claims that a concave mirror can produce both real and virtual images, whereas a convex mirror can produce only virtual images. Justify the statement with suitable reasoning.

51. A lens has a power of -2.5 D.

- Identify the type of lens.
- Calculate its focal length.
- State the nature of image formed when an object is placed in front of it.

52. (A) Complete the following ray diagram to show the formation of image:



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(B) Mention the nature, position and size of the image formed in this case.

(C) State the sign of the image distance in this case using the Cartesian sign convention.

53. (a)

(i) Draw a ray diagram to show the path of the refracted ray in each of the following cases:

- A ray of light incident on a concave lens parallel to its principal axis.
- A ray of light directed towards the principal focus of a concave lens.

(ii) A 4 cm tall object is placed perpendicular to the principal axis of a convex lens of focal length 24 cm. The distance of the object from the lens is 16 cm. Find the position and size of the image formed.

OR

(b)

(i) Draw a ray diagram to show the path of the reflected ray in each of the following cases:

- A ray of light incident on a convex mirror parallel to its principal axis.
- A ray of light directed towards the principal focus of a convex mirror.

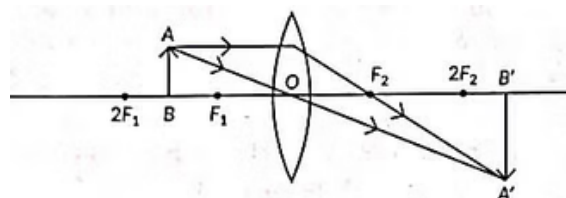
(ii) A 1.5 cm tall candle flame is placed perpendicular to the principal axis of a concave mirror of focal length 12 cm. If the distance of the flame from the pole of the mirror is 18 cm, use the mirror formula to determine the position and size of the image formed

Q54. Study the ray diagram given below and answer the questions that follow:

(a) Is the type of lens used converging or diverging?

(b) List three characteristics of the image formed.

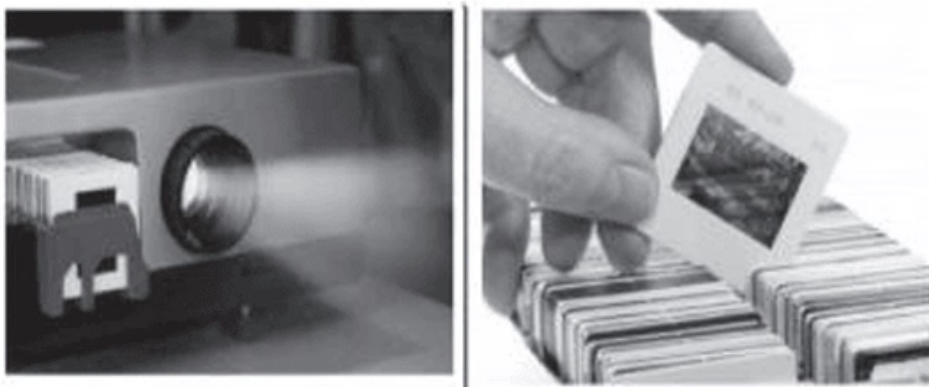
(c) In which position of the object will the magnification be -1 ?



PART 3 (Advanced Questions)

Case Study/Source Based Question (5 Marks)

55. Read the source below and answer the questions that follow:

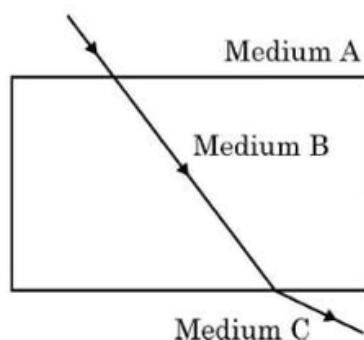


The images shown above are of a specialised slide projector. Slides are small transparencies mounted in sturdy frames that are ideally suited for magnification and projection because they possess very high resolution and excellent image quality. The projector contains a tray in which the slides are inserted in a particular orientation so that the audience can view enlarged erect images on the screen. For this reason, the slides are inserted upside down in the projector tray.

To show her students the images of insects that she investigated in the laboratory, Mrs. Iyer brought a slide projector. Her slide projector produced a 500 times enlarged and inverted image of a slide on a screen placed 10 m away.

1. Based on the information given above, identify the type of lens used in the slide projector.
2. If v represents image distance and u represents object distance, state the sign convention for both in this case with one reason.
3. A slide projector has a convex lens of focal length 20 cm. The slide is placed upside down at a distance of 21 cm from the lens. Calculate the distance at which the screen should be placed from the lens so that the image is obtained in clear focus.

Q56. (A) (a) Observe the following diagram and compare (i) speed of light and (ii) optical density of the three media A, B, and C. Also give justification for your answer of any one of the two cases in terms of refractive indices of A, B, and C.



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(b) Redraw the path of a ray of light through the three media, if the ray of light starting from medium A falls on the medium B:

- (i) Obliquely and the optical density of medium B is made more than that of A and C.
- (ii) The ray falls normally from medium A to medium B.

57. Analyse the following observation table showing variation of image distance (v) with object distance (u) in case of a convex lens and answer the questions that follow without doing any calculations:

Observation Number	Object distance (u) in cm	Image distance (v) in cm
1	-15	-60
2	-25	+100
3	-30	+60
4	-40	+40
5	-60	+30
6	-100	+25

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- Determine the focal length of the lens. Give reason for your answer.
- Find magnification of the image formed in Observation No. 3.
- The numerical value of magnifications in cases of observation 1 and 2 is same. List two differences in the images formed in these two cases.

58. Study the data given below showing the focal length of three concave mirrors A, B and C and the respective distances of objects placed in front of the mirrors :

Case	Mirror	Focal Length (cm)	Object Distance (cm)
1	A	20	45
2	B	15	30
3	C	30	20

- In which one of the above cases the mirror will form a diminished image of the object ? Justify your answer.
- List two properties of the image formed in case 2.
- (A) What is the nature and size of the image formed by mirror C? Draw ray diagram to justify your answer.
OR
- (B) An object is placed at a distance of 18 cm from the pole of a concave mirror of focal length 12 cm. Find the position of the image formed in this case.

59. The ability of a medium to refract light is expressed in terms of its optical density. Optical density has a definite connotation. It is not the same as mass density. On comparing two media, the one with the larger refractive index is optically denser than the other. The other medium with a lower refractive index is optically rarer. Also, the speed of light through a given medium is inversely proportional to its optical density.

- Determine the speed of light in diamond if the refractive index of diamond with respect to vacuum is 2.42. Speed of light in vacuum is 3×10^8 m/s.
- Refractive indices of glass, water and carbon disulphide are 1.5, 1.33 and 1.62 respectively. If a ray of light is incident in these media at the same angle (say θ), then write the increasing order of the angle of refraction in these media.
- (A) The speed of light in glass is 2×10^8 m/s and in water is 2.25×10^8 m/s.
 - Which one of the two is optically denser and why?
 - A ray of light is incident normally at the water-glass interface when it enters a thick glass container filled with water. What will happen to the path of the ray after entering the glass? Give reason.

OR

- (B) The absolute refractive indices of glass and water are $\frac{3}{2}$ and $\frac{4}{3}$, respectively. If the speed of light in glass is 2×10^8 m/s, calculate the speed of light in:
 - Vacuum
 - Water

60. Many optical instruments consist of a number of lenses. They are combined to increase the magnification and sharpness of the image. The net power (P) of the lenses placed in contact is given by the algebraic sum of the powers of the individual lenses $P_1, P_2, P_3 \dots$ as:

$$P = P_1 + P_2 + P_3 + \dots$$

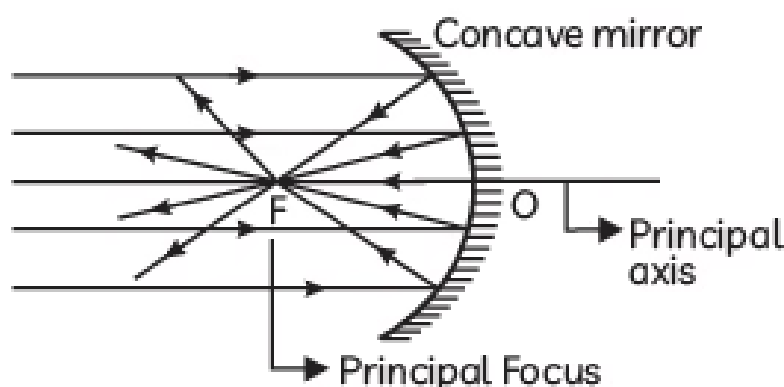
This is also termed as the simple additive property of the power of lenses, widely used to design lens systems of cameras, microscopes and telescopes. These lens systems can have a combination of convex lenses as well as concave lenses.

- What is the nature (convergent/divergent) of the combination of a convex lens of power $+4\text{ D}$ and a concave lens of power -2 D ?
- Calculate the focal length of a lens of power -2.5 D .
- Draw a ray diagram to show the nature and position of an image formed by a convex lens of power $+0.1\text{ D}$, when an object is placed at a distance of 20 cm from its optical centre.

OR

- How is a virtual image formed by a convex lens different from that formed by a concave lens? Under what conditions do a convex lens and a concave lens form virtual images?

61. Hold a concave mirror in your hand and direct its reflecting surface towards the Sun. Direct the light reflected by the mirror on to a white cardboard held close to the mirror. Move the cardboard back and forth gradually until you find a bright, sharp spot of light on the board. This spot of light is the image of the Sun on the sheet of paper, which is also termed as the “Principal Focus” of the concave mirror.



- List two applications of a concave mirror.
- If the distance between the mirror and the principal focus is 15 cm , find the radius of curvature of the mirror.
- Draw a ray diagram to show the type of image formed when an object is placed between the pole and focus of a concave mirror.
- An object 10 cm in size is placed at 100 cm in front of a concave mirror. If its image is formed at the same point where the object is located, find:
 - Focal length of the mirror
 - Magnification of the image formed with sign as per Cartesian sign convention.

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- Reflection by Plane Mirrors – Q1, Q2, Q3, Q4
- Refraction Through Lenses – Q5, Q8, Q9, Q13, Q34, Q39, Q41, Q44, Q47, Q48, Q51, Q52, Q53, Q54, Q55, Q57, Q60
- Reflection by Spherical Mirrors – Q6, Q7, Q10, Q11, Q12, Q14, Q15, Q16, Q17, Q18, Q19, Q28, Q29, Q30, Q31, Q33, Q36, Q37, Q38, Q42, Q43, Q45, Q49, Q50, Q58, Q61
- Refraction of Light – Q20, Q23, Q24, Q25, Q32, Q35, Q40, Q46, Q56, Q59
- Assertion & Reason Based Questions – Q21, Q22, Q23, Q24, Q25, Q26, Q27
- Concave Mirror – Q6, Q7, Q15, Q18, Q28, Q30, Q31, Q36, Q38, Q42, Q43, Q45, Q50, Q58, Q61
- Convex Mirror – Q10, Q11, Q14, Q16, Q17, Q37, Q49, Q50, Q52
- Convex Lens – Q5, Q8, Q9, Q34, Q39, Q41, Q44, Q47, Q48, Q52, Q53, Q54, Q55, Q57, Q60
- Concave Lens – Q13, Q47, Q51, Q53
- Magnification – Q14, Q18, Q28, Q29, Q30, Q42, Q49, Q57, Q61
- Mirror Formula & Numerical Problems – Q7, Q12, Q15, Q30, Q31, Q42, Q58, Q61
- Lens Formula & Numerical Problems – Q8, Q9, Q44, Q48, Q51, Q53, Q55, Q57, Q60
- Power of Lens – Q8, Q48, Q51, Q60
- Real and Virtual Images – Q6, Q10, Q11, Q13, Q15, Q16, Q28, Q29, Q30, Q31, Q36, Q39, Q41, Q50, Q60, Q61
- Applications of Mirrors and Lenses – Q14, Q17, Q26, Q37, Q49, Q52, Q55, Q61
- Competency & HOTS Based Questions – Q36, Q37, Q38, Q39, Q40, Q41, Q42, Q43, Q44, Q45, Q46, Q47, Q48, Q49, Q50, Q51, Q52, Q53, Q54, Q55, Q56, Q57, Q58, Q59, Q60, Q61



SOLUTIONS

MULTIPLE CHOICE QUESTIONS [MCQ's]

- | | | |
|--------|--------|--------|
| 1.(d) | 11.(a) | 21.(a) |
| 2.(b) | 12.(b) | 22.(a) |
| 3.(a) | 13.(c) | 23.(a) |
| 4.(b) | 14.(a) | 24.(a) |
| 5.(c) | 15.(d) | 25.(a) |
| 6.(b) | 16.(d) | 26.(c) |
| 7.(d) | 17.(b) | 27.(d) |
| 8.(c) | 18.(d) | |
| 9.(a) | 19.(a) | |
| 10.(d) | 20.(a) | |

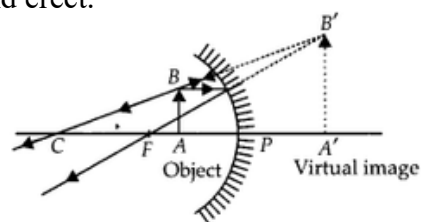
SHORT ANSWER TYPE QUESTIONS:

28. Negative sign of magnification indicates that the image is real and inverted. Since the image is real and inverted, the mirror is concave and magnification of -3 indicates that the image is magnified.

29. Positive value of the magnification indicates that image is virtual and erect.

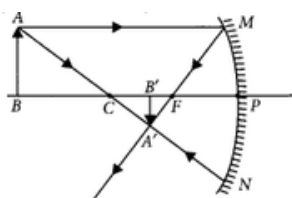
(i) Since the image is magnified, the mirror is concave.

(ii) The object is between pole and focus of the mirror as shown



30. (i) Concave mirror

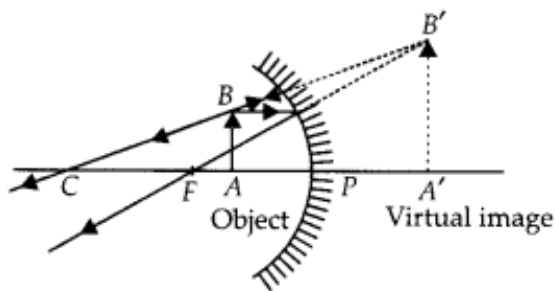
(ii) Object is placed beyond C.



31. Given, $f = -15$ cm, $u = -10$ cm.

Thus the object is placed between the principal focus and pole of the mirror.

- (a) The position of the image will be behind the mirror.
- (b) The size of the image will be highly enlarged.
- (c) The nature of the image will be virtual and erect



32. The refractive index of carbon disulphide is 1.63 means that light travels 1.63 times slower in carbon disulphide than in air or vacuum.

Calculation:

$$n = c/v$$

$$1.63 = (3 \times 10^8)/v$$

$$v = (3 \times 10^8)/1.63$$

$$v \approx 1.84 \times 10^8 \text{ m/s}$$

33. Given:

$$R = 50 \text{ cm}$$

Calculation:

$$f = R/2$$

$$f = 50/2 = 25 \text{ cm}$$

Type of mirror = Convex mirror

34. The beam of light passed undeviated because it was directed through the optical centre of the convex lens. A ray passing through the optical centre emerges without deviation.

Q35.

Given:

$$n = 1.5$$

$$c = 3 \times 10^8 \text{ m/s}$$

Calculation:

$$n = c/v$$

$$1.5 = (3 \times 10^8)/v$$

$$v = (3 \times 10^8)/1.5$$

$$v = 2 \times 10^8 \text{ m/s}$$

36. The first student is correct.

A concave mirror forms a virtual, erect, and enlarged image when the object is placed between the pole and focus. Therefore, all images formed by a concave mirror are not inverted.

37. A convex mirror is used as a rear-view mirror because it provides a wider field of view and forms virtual, erect, and diminished images.

38. The object is placed between the pole and focus of the concave mirror.

The image formed is virtual, erect, magnified, and behind the mirror.

39. No, I do not agree. A convex lens forms real and inverted images when the object is placed beyond its focal length.

40. Light bends towards the normal when entering glass from air because it moves from a rarer medium to a denser medium and slows down. It bends away from the normal when moving from glass to air because it moves from a denser medium to a rarer medium and speeds up.

41. The image cannot be obtained on a screen because it is a virtual image formed by apparent intersection of rays. The rays do not actually meet.

LONG ANSWER TYPE QUESTIONS:

42. Given:

$$v = -36 \text{ cm}$$

$$m = -3$$

Calculation:

$$m = -v/u$$

$$-3 = -(-36)/u$$

$$-3 = 36/u$$

$$u = -12 \text{ cm}$$

Using mirror formula:

$$1/f = 1/v + 1/u$$

$$1/f = 1/(-36) + 1/(-12)$$

$$1/f = (-1 - 3)/36$$

$$1/f = -4/36$$

$$1/f = -1/9$$

$$f = -9 \text{ cm}$$

Answer:

Object distance, $u = -12 \text{ cm}$

Focal length, $f = -9 \text{ cm}$

43. Image formed is:

- Real and inverted
- Enlarged
- Beyond the centre of curvature

If the object moves towards the focus, the image moves farther away and becomes more enlarged.

44. Image distance, $v = +80 \text{ cm}$

$$\text{Magnification, } m = -2$$

Because the image formed by a convex lens on a screen is real and inverted.

Using magnification formula:

$$m = v/u$$

$$-2 = 80/u$$

$$u = -40 \text{ cm}$$

Now using lens formula:

$$1/f = 1/v - 1/u$$

$$1/f = 1/80 - 1/(-40)$$

$$1/f = 1/80 + 1/40$$

$$1/f = 1/80 + 2/80$$

$$1/f = 3/80$$

Therefore,

$$f = 80/3 = 26.7 \text{ cm}$$

Final Answer: The focal length of the convex lens is 26.7 cm.

45. The object should be placed between the pole and focus of the concave mirror, i.e., at a distance less than 20 cm from the mirror.

Reason: When the object is placed between the pole and focus, the concave mirror forms a virtual, erect, and highly enlarged image behind the mirror.

46. When light travels from glass to air, it goes from a denser medium to a rarer medium. Its speed increases, so the ray bends away from the normal. This bending of light is called refraction.

47. When parallel rays fall on a convex lens, the rays converge at the principal focus.

When parallel rays fall on a concave lens, the rays diverge and appear to come from the principal focus.

Image formed by:

- Convex lens: Can form real and inverted images.
- Concave lens: Always forms virtual, erect, and diminished images.

48. Given:

Object distance, $u = -24 \text{ cm}$

Image distance, $v = -8 \text{ cm}$

Using lens formula:

$$1/f = 1/v - 1/u$$

$$1/f = 1/(-8) - 1/(-24)$$

$$1/f = -1/8 + 1/24$$

$$1/f = (-3 + 1)/24$$

$$1/f = -2/24$$

$$1/f = -1/12$$

Therefore,

$$f = -12 \text{ cm}$$

Now, power of lens:

$$P = 100/f \text{ in cm}$$

$$P = 100/-12$$

$$P = -8.33 \text{ D}$$

Final Answer:

Focal length = -12 cm

Power = -8.33 D

49. The mirror used is a convex mirror.

Reason: The magnification is positive and less than 1 (+0.25), which is the property of a convex mirror.

It is preferred over a plane mirror because:

- it provides a wider field of view,
- forms erect and diminished images,
- helps the driver see more traffic behind.

Q50. The statement is correct.

A concave mirror:

- forms real and inverted images when the object is beyond the focus,
- forms virtual, erect, and enlarged images when the object is placed between the pole and focus.

A convex mirror always forms virtual, erect, and diminished images irrespective of the object position.

51. Given:

Power, $P = -2.5 \text{ D}$

(i) Type of lens:

Concave lens

(ii) Calculation of focal length:

$P = 1/f(\text{in metre})$

$-2.5 = 1/f$

$f = 1/(-2.5)$

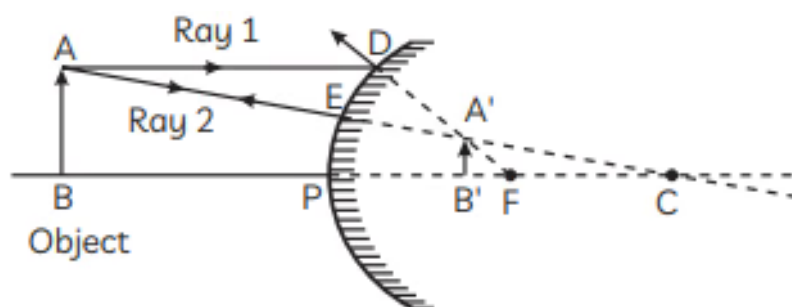
$f = -0.4 \text{ m}$

$f = -40 \text{ cm}$

(iii) Nature of image formed:

The image formed is always virtual, erect, and diminished.

52.

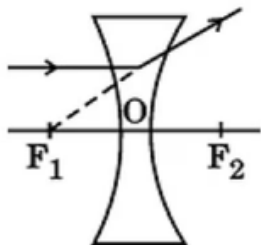


(B) The image of object obtained in the convex mirror is erect and diminished. This is because a convex mirror always forms a virtual, erect and diminished image of an object.

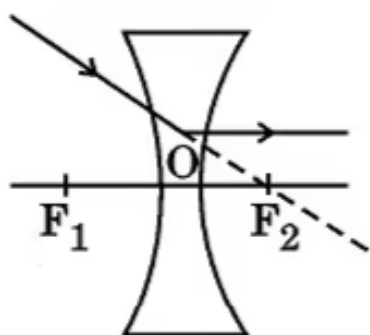
(C) The image distance is positive. This is because, the image is formed is behind the mirror.

53. Ans: (a) (i)

(1) Parallel to its principal axis



(2) is directed towards its principal focus



(ii) Given $u = -16 \text{ cm}$, $f = +24 \text{ cm}$, $h = 4 \text{ cm}$

Lens Formula:

$$1/f = 1/v - 1/u$$

$$1/24 = 1/v + 1/16 \Rightarrow 1/v = 1/24 - 1/16$$

$$\Rightarrow 1/v = (2 - 3)/48$$

$$\Rightarrow 1/v = -1/48 \Rightarrow v = -48 \text{ cm}$$

Magnification: $m = v/u$

$$= (-48)/(-16) = 3$$

Image Height:

$$h' = m \times h$$

$$= 3 \times 4$$

$$= 12 \text{ cm}$$

54. (a) We have used a converging lens.

(b) The characteristics of the image formed:

(i) It is real. (ii) It is inverted (iii) It is enlarged.

(c) We get the magnification of object, $m = -1$ at the position $2F_1$.

CASE STUDY/SOURCE BASED QUESTION:

55. 1. Lens used:

A convex lens is used because it forms a real, inverted and enlarged image on the screen.

2. Sign convention:

u = negative because the object is on the left side of the lens.

v = positive because the real image is formed on the right side of the lens.

3. Calculation:

Given: $f = +20 \text{ cm}$, $u = -21 \text{ cm}$

Using lens formula:

$$1/f = 1/v - 1/u$$

$$1/20 = 1/v - 1/(-21)$$

$$1/20 = 1/v + 1/21$$

$$1/v = 1/20 - 1/21$$

$$1/v = 1/420$$

$$v = 420 \text{ cm}$$

56. a) From the diagram, the ray bends towards the normal when it enters medium B from medium A.

So, medium B is optically denser than medium A.

When the ray goes from medium B to medium C, it bends away from the normal.

So, medium C is optically rarer than medium B.

Therefore:

Optical density:

$$B > A \text{ and } B > C$$

Speed of light:

Speed of light is least in medium B.

Refractive index:

$$n_B > n_A \text{ and } n_B > n_C$$

(b)(i)

If medium B is made optically denser than both A and C, then the ray will bend towards the normal while entering from A to B, and it will bend away from the normal while going from B to C.

(b)(ii) If the ray falls normally from medium A to B, it will pass undeviated.

Reason: At normal incidence, the angle of incidence is 0° , so the ray does not bend during refraction.

57. (a) Focal Length: From observation 4, $u = -40$ cm, $v = +40$ cm.

Lens formula: $1/f = 1/v - 1/u$

$$= 1/40 - 1/(-40)$$

$$= 1/40 + 1/40$$

$$= 2/40 = 1/20$$

$\therefore f = 20$ cm (positive, convex lens)

Reason: The data is consistent with a convex lens of focal length 20 cm, as verified across observations.

(b) Magnification for Observation 3: $u = -30$ cm, $v = +60$ cm.

$$m = v/u$$

$$= 60/(-30) = -2$$

The magnification is -2 (inverted, twice the size).

(c) Differences:

- Observation 1: $u = -15$ cm, $v = -60$ cm.

$$m = v/u = -60/(-15) = 4$$

(positive, virtual, erect image, same side)

- Observation 2: $u = -25$ cm, $v = +100$ cm.

$$m = v/u = 100/(-25) = -4$$

(negative, real, inverted image, opposite side)

Differences:

1. Observation 1: Virtual, erect image; Observation 2: Real, inverted image.

2. Observation 1: Image on same side as object; Observation 2: Image on opposite side.

58. (i) Diminished image

Mirror A will form a diminished image.

For mirror A:

Focal length, $f = 20$ cm

Centre of curvature, $2f = 40$ cm

Object distance, $u = 45$ cm

Since the object is placed beyond the centre of curvature, the image formed will be real, inverted and diminished.

ii) Properties of image in case 2

For mirror B: Focal length, $f = 15$ cm

Centre of curvature, $2f = 30$ cm

Object distance, $u = 30$ cm

So, the object is placed at the centre of curvature.

Two properties of the image:

1. Image is real and inverted.

2. Image is of the same size as the object.

(iii) (A) Image formed by mirror C

Focal length, $f = 30$ cm

Object distance, $u = 20$ cm

Since $u < f$, the object is placed between the pole (P) and focus (F) of the concave mirror.

So, the image formed is: Virtual, Erect and Magnified

Formed behind the mirror

OR

(iii) (B) Numerical

Given:

$$u = -18 \text{ cm}$$

$$f = -12 \text{ cm}$$

Using mirror formula:

$$1/f = 1/v + 1/u$$

$$1/-12 = 1/v + 1/-18$$

$$-1/12 = 1/v - 1/18$$

$$1/v = -1/12 + 1/18$$

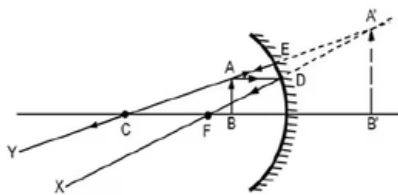
$$1/v = (-3 + 2)/36$$

$$1/v = -1/36$$

Therefore,

$$v = -36 \text{ cm}$$

Final Answer: The image is formed 36 cm in front of the mirror.



59. (i) Given: $n = 2.42$, $c = 3 \times 10^8 \text{ m/s}$

$$n = c/v$$

$$2.42 = 3 \times 10^8 / v$$

$$v = 1.24 \times 10^8 \text{ m/s}$$

(ii) Higher refractive index $\rightarrow$ smaller angle of refraction.

Refractive index order:

Carbon disulphide $>$ Glass $>$ Water

So, increasing order of angle of refraction:

$r_{\text{carbon disulphide}} < r_{\text{glass}} < r_{\text{water}}$

(iii) (A)

(a) Glass is optically denser because light travels slower in glass.

Answer: Glass

(b) The ray will pass undeviated because it is incident normally. Only speed changes.

OR (B)

Given: $n_g = 3/2$, $n_w = 4/3$, speed in glass $= 2 \times 10^8 \text{ m/s}$

$$(i) \frac{3}{2} = c / 2 \times 10^8$$

$$c = 3 \times 10^8 \text{ m/s}$$

$$(ii) \frac{4}{3} = 3 \times 10^8 / v$$

$$v = 2.25 \times 10^8 \text{ m/s}$$

60. (a) Given: $P_1 = 4 \text{ D}$, $P_2 = -2 \text{ D}$

$$P = P_1 + P_2$$

$$= 4 \text{ D} - 2 \text{ D}$$

$$= 2 \text{ D} \text{ So the lens is convergent in nature.}$$

$$(b) P = -2.5 \text{ D}$$

$$P = 100/f (\text{cm})$$

$$\Rightarrow -2.5 = 100/f$$

$$\Rightarrow f = -40 \text{ cm}$$

$$(c) P = 0.1 \text{ D}, u = -20 \text{ cm}$$

$$P = 100/f \text{ (cm)}$$

$$\Rightarrow 0.1 = 100/f$$

$$\Rightarrow f = 1000 \text{ cm}$$

$$1/f = 1/v - 1/u$$

$$1/1000 = 1/v - 1/(-20)$$

$$1/1000 = 1/v + 1/20$$

Step 3: Solve for v

$$1/v = 1/1000 - 1/20$$

Taking LCM (1000):

$$1/v = (1 - 50)/1000$$

$$1/v = -49/1000$$

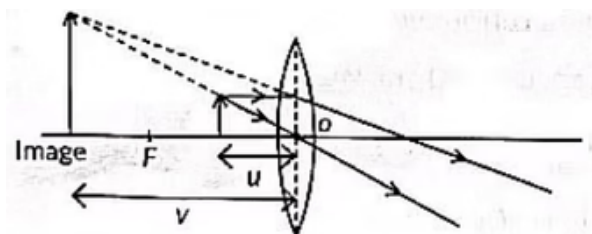
$$v = -1000/49$$

$$v \approx -20.4 \text{ cm}$$

Nature of Image:

Since v is negative, the image is formed on the same side as the object.

Therefore, the image is virtual and erect.



OR

(c) Virtual images formed by convex and concave lenses differ in several ways:

- Convex Lens: Forms a virtual image when the object is within the focal length. The image is erect, magnified and on the same side as the object.
- Concave Lens: Always forms a virtual image. The image is erect, diminished and on the same side as the object.

61. (A) Applications of concave mirrors:

- (1) A concave mirror is used as a shaving mirror when the face is placed close to it so that it is within its focus and we get an erect and magnified image of the face.
- (2) Doctors use concave mirrors as head mirrors to concentrate parallel rays of light at its focus, which enables them to examine body parts such as eyes, throat, etc.

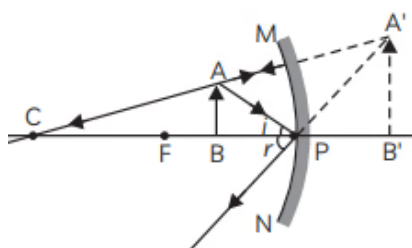
(B) Given, $f = 15 \text{ cm}$

We know for a mirror, $R = 2f$

$$R = 2 \times 15 \text{ cm}$$

$$R = 30 \text{ cm}$$

(C)



(D)(i) We can use the mirror formula to find the focal length of the mirror: $1/f = 1/v + 1/u$

where, f is the focal length of the mirror, v is the image distance and u is the object distance. Since the image is formed at the same point as the object,

$$v = u = -100 \text{ cm}$$

(Distances to the left of the mirror are negative.)

Substituting the values, we get:

$$1/f = 1/(-100) + 1/(-100)$$

$$1/f = -2/100$$

$$f = -50 \text{ cm}$$

So the focal length of the mirror is -50 cm .

(Negative sign indicates that it is a concave mirror.)

(ii) The magnification of the image is:

$$m = -v/u$$

where, m is the magnification of the image.

Substituting the values, we get:

$$m = -(-100/-100)$$

$$m = -1$$

So the magnification of the image is -1 .

(Negative sign indicates that the image is real and inverted.)

Class 10th
Phodenge 🔥

